

Role of Telemetry Monitoring in the Non-Intensive Care Unit

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To determine the outcomes of patients admitted to a non-intensive care telemetry unit and to assess the role of telemetry for guiding patient management decisions, data from 2,240 patients admitted to a telemetry unit were collected prospectively during 7 months. Physicians recorded the outcomes (intensive care unit transfer and mortality) and assessed whether telemetry assisted in guiding patient management. Indications for admission to the telemetry unit included chest pain syndromes (55%), arrhythmias (14%), heart failure (12%), and syncope (10%). Telemetry led to direct modifications in management in 156 patients (7%; 95% confidence interval [CI] 5.9% to 8%). Telemetry was perceived as useful but did not alter management for 127 patients (5.7%; 95% CI 4.7% to 6.6%). Two hundred

forty-one patients were transferred to an intensive care unit from the telemetry unit (10.8%; 95% CI 9.5% to 12%). Nineteen patients (0.8% of all admissions; 95% CI 0.5% to 1.2%) were transferred because of an arrhythmia identified by telemetry. Routine transfer after cardiac revascularization or surgery accounted for 134 transfers; clinical deterioration accounted for 88 transfers. There were 20 deaths in the unit (0.9%; 95% CI 0.5% to 1.3%); 4 of the 20 deaths occurred while patients were being monitored. The role of telemetry in guiding patient management may be overestimated by physicians, since it detected significant arrhythmias that led to change in medications or urgent interventions in a small fraction of patients.

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Management of patients with suspected myocardial ischemia in the emergency department or intensive care units has been addressed in numerous studies.¹⁻¹⁰ Despite the fact that cardiac monitoring in non-intensive care units is frequently used and that it adds to the cost of health care, only 2 studies have been published regarding patients admitted to such units. Lipskis et al¹¹ reported that significant arrhythmias occurred only in patients with known or suspected coronary artery disease or in those with previously documented arrhythmia. We¹² previously reported that telemetry identified an arrhythmia that resulted in intensive care unit transfer in only 5 of 467 patients (1.1%). Telemetry identified a terminal rhythm in 3 patients but did not alter clinical decisions.¹² It is assumed that telemetry monitoring enhances medical care in non-intensive care unit wards. However, there are no studies documenting the effectiveness of telemetry in guiding decisions regarding patient care. In this study we describe the outcomes of patients admitted to a non-intensive care telemetry unit and examine the role of telemetry for guiding patient management decisions.

METHODS

Practice setting: Henry Ford Hospital is a 900-bed primary and tertiary care facility, and the principal hospital for a 1,100-member medical group. Beds with telemetry monitoring are located in intensive care units,

step-down units, recovery room, emergency room, and a 72-bed telemetry unit.

The telemetry unit has the capacity to monitor 36 of the 72-bed complement with portable monitors. Patients are admitted to this unit from surgical and medical intensive care units, recovery room, emergency room, outpatient clinics, and from home. Heart rate and rhythm are recorded continuously with a Hewlett-Packard 78720 A SDN system (Rockville, Maryland); this system allows automatic recognition of arrhythmias. Technicians trained in arrhythmia recognition watch the monitor 24 hours/day and routinely report to the medical staff. The nurse-to-patient ratio is similar to that of the general medicine wards. Nursing personnel are certified and trained in advanced cardiac life support to recognize and manage arrhythmias. Physicians on this unit include house staff, cardiology fellows, and cardiology faculty. Care of patients admitted to the telemetry unit is the responsibility of these physician teams. No attempt was made to alter the decisions or behavior of physicians working on the unit.

Patient selection: The selection criteria were chosen in order to evaluate the role of telemetry for patients likely to be seen by internists and cardiologists and for whom the role of telemetry has not been defined. These patients may represent a significant number of admissions per year and may consume substantial health care resources. Patients were considered eligible for the study if they were admitted to the telemetry unit. The exclusion criteria were admission to a surgical team, and transfer from an intensive care unit or recovery room; thus, we excluded patients in whom cardiac monitoring was deemed essential. The excluded group comprised patients recovering from cardiac revascularization, patients who had

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sustained a recent myocardial infarction, or those in whom significant arrhythmias had initially been controlled in an intensive care setting.

Patients were followed until discharge from the telemetry unit. Mortality and intensive care unit transfer for eligible patients who were not followed prospectively were obtained at the end of the study. The study was performed between June 30, 1993, and January 31, 1994. The study was approved by the Human Rights Committee at Henry Ford Hospital.

Baseline information: Clinical information was collected prospectively by the physician teams working on the telemetry unit. The admitting house staff recorded the age, gender, race, historical information, and site from which the patient was admitted. History of coronary artery disease was defined as prior typical angina, myocardial infarction, percutaneous transluminal angioplasty, coronary artery bypass surgery, positive treadmill stress test, or cardiac catheterization with $\geq 50\%$ stenosis in ≥ 1 vessel. History of stroke or transient ischemic attacks, diabetes mellitus, congestive heart failure, hypertension, and smoking status were also documented. The use of intravenous inotropic drugs and resuscitation code status were recorded.

Only 1 principal reason for admission to telemetry monitoring was recorded; this was coded by the resident using the following hierarchy: (1) chest pain syndrome, (2) documented arrhythmia, (3) congestive heart failure, (4) syncope or lightheadedness, (5) coronary artery disease awaiting revascularization, and (6) other reason. The number of days and complications after myocardial infarction were noted in patients who were admitted from other institutions.

American College of Cardiology classification for the use of telemetry: Using a standardized form, 1 author classified each patient according to the American College of Cardiology guidelines for in-hospital cardiac monitoring for detection of arrhythmia.¹³ A second author then reviewed the first classification and consensus was achieved on disagreements. In our prior study, the interrater agreement was excellent ($\kappa = 0.8$).¹² The American College of Cardiology guidelines were tabulated by an expert consensus panel and are summarized as follows: class I, all patients at significant risk for immediate life-threatening arrhythmia; class II, patients in whom monitoring may be beneficial but is not deemed essential; and class III, patients in whom cardiac monitoring is generally not indicated as the risk of serious arrhythmia, or likelihood of therapeutic benefit is low.

Role of telemetry in guiding patient management: The cardiology faculty or fellows reviewed the telemetry information daily and documented the rhythm in a standardized form. The role of telemetry findings, either normal or abnormal, in guiding patient management was assessed by asking these physicians open-ended questions. They were asked to determine if: (1) it helped them guide medical therapy (started, stopped, or changed medications); (2) it helped them guide diagnostic studies (obtained or withheld a study); and (3) it helped them decide to transfer the patient to an intensive care unit.

TABLE I Utility of Telemetry in Guiding Patient Management as Perceived by Cardiology Faculty (n = 2,240)

	No. (%)
1. Telemetry monitoring was not helpful	1,760 (78.6)
2. Telemetry monitoring perceived as helpful	
Management not modified	127 (5.7)
Monitoring for rhythm disturbance in patient	
At potential risk and for surveillance only	58 (2.6)
(e.g., prior arrhythmia, syncope, chest pain)	
Given antiarrhythmic drug to assess its effect or arrhythmogenicity (e.g., amiodarone initiated)	40 (1.8)
Postcardiac intervention	
(e.g., pacemaker, electrophysiologic testing)	29 (1.3)
Management modified	156 (7)
Arrhythmia detection led to:	
Medication change*	112 (5)
Cardiac intervention	25 (1.1)
(e.g., pacemaker, cardioversion, resuscitation)	
Transfer to intensive care unit	19 (0.8)
3. Telemetry monitoring unknown	197 (8.8)

*Urgent medical intervention with adenosine was required in 4 patients with supraventricular tachycardia.

No a priori specific criteria of usefulness of telemetry were required. If patients' cardiologists determined that telemetry guided patient management, these forms were reviewed at the end of the study by 4 of the authors to achieve consensus. The categories described in Table I (section 2) were created at the end of the study using a Delphi technique¹⁴; patients were thus divided into whether the management was modified or not. The Delphi technique is a way to achieve consensus. Briefly, the open-ended responses were grouped into "common themes" by 2 authors. For example, in several patients a dobutamine infusion was performed when a tachyarrhythmia developed, and in other patients a β blocker was added when a supraventricular arrhythmia was not controlled with digoxin. These 2 groups had a common theme: The management was modified and it led to a medication change (Table I). The responses and the common themes were then independently reviewed by a third author and modified accordingly. The process was repeated until consensus was achieved.

Outcomes: Two clinical outcomes were measured: transfer to an intensive care unit and mortality in the telemetry unit. The cardiology staff wrote a brief description of the events in relation to intensive care unit transfer or death. This description was reviewed by 2 authors and disagreement between them was resolved by a third cardiologist. The interrater agreement for the reason for transfer was 86%. Intensive care unit transfers were categorized as (1) transfer resulting from arrhythmia detected by telemetry, or (2) transfer independent of telemetry findings (i.e., clinical deterioration or routine transfer after cardiac revascularization or surgery) (Table II).

Statistical methods: The expected rate of intensive care unit transfers was estimated to be 10%.¹² A sample size of 2,212 subjects is needed to obtain 95% confidence intervals (CIs), with a 5% width for this point estimate ($\alpha = 0.05$, $\beta = 0.80$). Descriptive statistics, chi-square, and 95% CIs were used where appropriate.¹⁵ All 2,240 eligible patients were included in the analyses.

TABLE II Indications for Intensive Care Unit Transfers, and Mortality After Intensive Care Unit Transfer (n = 241)

Indication for Transfer	Intensive Care Unit Transfer No. (%)	Mortality No. (%)
Guided by telemetry findings		
Transfer due to cardiac arrhythmia*	19 (7.9)	2/19 (10.5)
Atrial fibrillation or flutter	5 (2.1)	
Sinus bradycardia and hypotension	2 (0.8)	
Ventricular arrhythmia (torsades, sustained or wide complex tachycardia)	5 (2.1)	
Atrioventricular blocks (3rd degree, 2nd degree type II, junctional)	7 (2.9)	
Independent of telemetry findings		
Transfer due to clinical deterioration	88 (36.5)	17/88 (19.3)
Angina	30 (12)	
Heart failure	22 (9.1)	
Myocardial infarction	11 (4.6)	
Pulmonary	10 (4.1)	
Gastrointestinal	4 (1.7)	
Other†	7 (2.9)	
Cardiac or respiratory arrest	4 (1.7)	
Transfer after intervention	134 (55.6)	2/134 (1.5)
Coronary artery bypass grafting	45 (19)	
Percutaneous angioplasty	70 (29)	
Other cardiovascular surgery	13 (5.4)	
General surgery	6 (2.5)	
Total	241 (100)	21/241 (8.7)

*Arrhythmia was the sole indication for transfer of 16 of 19 patients. The patients with atrial fibrillation or flutter required cardioversion or overdrive pacing. Of these 19 patients, 10 had a rhythm abnormality on admission (supraventricular arrhythmia, torsades de pointes, digitalis toxicity, pacemaker malfunction); 4 were initially admitted with chest pain, and 5 had various conditions including congestive heart failure, endocarditis, or syncope.

†Abdominal pain with an abdominal aortic aneurysm, stroke, sepsis, cardiac tamponade, and vasovagal syncope in 1 patient after postmyocardial infarction.

RESULTS

Assessment of the sample: During the study period, 2,240 eligible patients were admitted to the non-intensive care telemetry unit. Information was collected prospectively in 2,036 (91%). An analysis of why data were not collected prospectively in 204 eligible patients (9% of the cohort) revealed 3 causes: house staff or cardiologist error, patient discharge within 12 hours, or lost forms.

Mortality in the telemetry unit occurred in 20 patients (0.9%; 95% CI 0.5% to 1.3%); it was 4.9% (10 of 204) in patients who were not followed, and 0.5% (10 of 2,036) in patients who were followed prospectively ($p < 0.001$). Intensive care unit transfers occurred in 241 patients (10.8%; 95% CI 9.5% to 12%); it was 9.3% (19 of 204) in patients who were not followed, and 11% (222 of 2,036) in patients who were followed prospectively ($p = 0.55$).

Patient characteristics and telemetry findings: Patient characteristics are listed in Table III. Most patients were admitted with chest pain syndromes (55%), followed by arrhythmia (14%), heart failure (12%), and syncope or lightheadedness (10%). Most patients (61%) were classified as class I, 38% as class II, and 1% as class III according to the American College of Cardiology guidelines for the use of telemetry. During follow-up, 5.9% of patients admitted with chest pain syndromes proved to have a myocardial infarction.

During hospitalization, telemetry showed 612 abnormal findings in 536 of 2,240 patients (24%; 95% CI 22.2% to 25.7%). Abnormal findings consisted of 272 episodes (12%) of supraventricular arrhythmia (atrial fibrillation, flutter, paroxysmal atrial tachycardia), 121 episodes (5.4%) of sinus bradycardia or tachycardia (heart rate < 50 or > 100 beats/min), 30 episodes (1.3%) of atrioventricular block (3rd degree, and 2nd degree type II), 51 of nonsustained ventricular tachycardia (2.3%), and 133 of other rhythm abnormalities (5.9%) (2nd degree type I, bundle branch block, junctional, paced, and so forth). Other ventricular arrhythmias were detected in 9 patients (0.4%), of whom 1 died (ventricular fibrillation). Five patients were transferred to the intensive care unit (sustained ventricular tachycardia, torsades de pointes, wide complex tachycardia [Table II]), and 3 remained on telemetry. Of the 3 patients who remained on telemetry, 2 had Wolff-Parkinson-White syndrome (1 underwent catheter ablation and the other was controlled with medical therapy); the third patient had an automatic internal cardiac defibrillator which fired repetitively for episodes of ventricular tachycardia.

Role of telemetry in guiding patient management:

Physicians perceived telemetry as helpful in 283 of 2,240 patients (12.6%; 95% CI 11% to 14%) (Tables I and IV). Telemetry led to direct modifications in management in 156 patients (7%; 95% CI 5.9% to 8%). In these patients, telemetry guided medical therapy (e.g., dobutamine was discontinued when a tachyarrhythmia developed), cardiac interventions (e.g., overdrive pacing decided for uncontrolled atrial flutter), or intensive care unit transfers (e.g., ventricular arrhythmias) (Table I).

Telemetry was perceived as useful but did not alter management in 127 of 2,240 patients (5.7%; 95% CI 4.7% to 6.6%) (Table I). In most patients, it was used as reassurance that no arrhythmias were identified, or to detect proarrhythmic effects of medications, or to monitor the function of a recently inserted pacemaker.

Physicians believed that telemetry did not alter the care of their remaining 1,760 patients (78.6%; 95% CI 77% to 80%) (Table I). Patients in this group did not have major arrhythmic events.

Physicians perceived telemetry as helpful in 51% of patients admitted with arrhythmias (Table IV), in 16% of patients with syncope, and in 4.4% of patients with chest pain syndromes.

Outcomes: INTENSIVE CARE UNIT TRANSFER: During the study period, 10.8% of patients (241 of 2,240) were transferred to an intensive care unit (95% CI 9.5% to 12%) (Table II). Transfers due to severe arrhythmia detected by telemetry occurred in 19 of these patients

(7.9%; 95% CI 4.5% to 11.3%, or 0.8% of all admissions; 95% CI 0.5% to 1.2%). Severe arrhythmia detected by telemetry occurred in 3.2% of patients admitted with arrhythmia (Table IV), 1.1% of patients admitted with heart failure, and in <0.8% of patients admitted with chest pain, syncope, or various other conditions. None of the patients awaiting cardiac revascularization sustained a severe arrhythmia (Table IV).

Transfers independent of telemetry findings were due to clinical deterioration in 88 patients (36.5%; 95% CI 30% to 43%) or to routine transfer after cardiac revascularization or surgery in 134 patients (55.6%; 95% CI 49% to 62%) (Table II). Transfer due to clinical deterioration occurred more frequently in patients initially admitted with heart failure (8.3%), patients awaiting cardiac revascularization (10.8%), and patients recovering from a myocardial infarction (7.9%) (Table IV). Twenty-one of the 241 patients (8.7%) transferred to an intensive care unit subsequently died (description next) (Table II).

MORTALITY: Twenty patients died on the telemetry unit (0.9%; 95% CI 0.5% to 1.3%) (Table IV). The reason for initial monitoring had been addressed or resolved and telemetry was discontinued before death in 16 of the 20 patients; the remaining 4 patients died shortly after admission, 2 from aspiration and 2 from sudden cardiac ischemia. Ten of the 20 patients had severe underlying conditions and requested not to be resuscitated.

An additional 21 patients died after transfer to an intensive care unit (Table II). Two of the patients who were routinely transferred died after complications of cardiac revascularization procedures (2 of 134, 1.5%). Two of the 19 patients (10.5%) who were transferred because of cardiac arrhythmia died because of sepsis or congestive heart failure. Of the patients who were transferred due to clinical deterioration, 17 died (17 of 88, 19.3%) due to cardiac ($n = 11$, progressive heart failure, myocardial infarction) or medical problems ($n = 6$, sepsis, stroke, abdominal catastrophe) (Table II).

DISCUSSION

In this study, we show that the role of telemetry in guiding patient management decisions may be smaller than expected. Nineteen patients (0.8% of all admis-

TABLE III Patient Characteristics ($n = 2,240$)

Variable	Value*
Age (yr) (mean $\pm$ SD)	63.5 $\pm$ 15.4
Length of stay on telemetry unit (d) (mean $\pm$ SD)	3.5 $\pm$ 3.1
Sex	
Men	1,037 (46)
Women	1,021 (46)
Race	
Black	1,150 (51)
White	842 (38)
Other	69 (3)
Admitted from:	
Emergency room	1,519 (68)
Other hospital units or institutions	314 (14)
Outpatient clinic or home	169 (8)
Medical history	
Hypertension	1,109 (50)
Coronary artery disease	673 (30)
Heart failure	444 (20)
Diabetes mellitus	399 (18)
Current smoker	206 (9)
Stroke or transient ischemic attacks	137 (6)
Do not resuscitate status	52 (2.3)
Use of intravenous inotropic drugs	16 (0.7)
Reason for admission	
Chest pain syndromes	1,225 (55)
Arrhythmia [†]	315 (14)
Congestive heart failure	277 (12)
Syncope or lightheadedness	227 (10)
Other [‡]	121 (5.4)
Coronary disease awaiting revascularization	37 (1.7)
Postmyocardial infarction	38 (1.7)
American College of Cardiology classification for the use of telemetry	
Class I (indicated in most)	1,356 (61)
Class II (indicated in some)	859 (38)
Class III (not indicated)	25 (1)

*Some data were unknown in up to 10.6% of patients.

[†]Supraventricular (176 of 315, 56%; atrial fibrillation, flutter, supraventricular tachycardia), sinus (15, 4.8%; bradycardia or tachycardia), nonsustained ventricular tachycardia (10, 3.2%), sustained ventricular tachycardia (6, 1.9%), atrioventricular block (23, 7.3%; 2nd and 3rd degree), other (29, 9.2%, e.g., junctional rhythm, bifascicular block, sick sinus syndrome, bigeminy, and so forth), and unknown (56, 17.8%). The arrhythmia was previously known in 106 of 315 patients (34%).

[‡]Drug intoxication, respiratory conditions (asthma, chronic obstructive lung disease, pneumothorax, pneumonia, pulmonary embolism), chest trauma, gastrointestinal bleeding, sepsis, seizures, hypertension, pericarditis, stroke, and so forth.

Values are expressed as number (%) unless otherwise noted.

TABLE IV Outcomes and Utility of Telemetry

Reason for Admission	Mortality (%)	Intensive Care Unit Transfer			Telemetry Perceived as Helpful* (%)	No. (%)
		Due to Cardiac Arrhythmia (%)	Due to Clinical Deterioration (%)	After Intervention (%)		
Chest pain syndromes	0.4	0.3	3.4	6.3	4.4	1,225 (100)
Arrhythmia	1	3.2	3.2	1.9	51	315 (100)
Congestive heart failure	2.9	1.1	8.3	3.3	7.6	277 (100)
Syncope or lightheadedness	0	0.4	0.9	1.3	16	227 (100)
Other	2.5	0.8	4.1	5.8	8.3	121 (100)
Coronary disease awaiting revascularization	0	0	10.8	49	0	37 (100)
Postmyocardial infarction	2.6	0	7.9	37	5.3	38 (100)
Total (%)	20 (0.9)	19 (0.8)	88 (3.9)	134 (6)	283 (12.6)	2,240 (100)

*As perceived by patient's cardiologists. In these examples, telemetry data guided therapy or guided use of diagnostic tests or assisted in the decision to transfer a patient to an intensive care unit.

sions) were transferred to an intensive care unit because of an arrhythmia identified by telemetry. The remaining 222 transfers were unrelated to the telemetry findings. Overall, telemetry was perceived as helpful in 12.6% of patients admitted to the unit.

Cardiac monitoring in non-intensive care settings is increasingly being used to decrease the demand for beds in intensive care units.¹⁶ However, little is known about how telemetry is used in this setting and whether its use is appropriate or effective.^{11,12} We previously reported on the outcomes of patients admitted to a non-intensive care telemetry unit.¹² In the present study we extended our observations in 3 areas. First, we monitored mortality and intensive care unit transfers in a larger cohort of patients (2,240 vs 467). Second, we examined the role of telemetry on the decision to transfer patients to an intensive care unit. Third, and more important, we evaluated whether telemetry affected decision making in more subtle ways by guiding patient management.

Intensive care unit transfers occurred in 10.8% of our current cohort (241 of 2,240) versus 16% in our prior study. The indications for intensive care unit transfers were similar for both reports: arrhythmia detection (7.9% in the current vs 6.7% in the prior study), clinical deterioration (36.5% vs 41%), or after an intervention (55.6% vs 53%) ($p = 0.8$). Death occurred in 0.9% of patients in the current study. The higher mortality in the prior report (1.9%) could be explained by the fact that in the prior report patients were followed during the entire hospitalization, whereas in the current study patients were followed until discharge or transfer.

Does telemetry have any impact on outcomes? No randomized controlled trial has ever studied this question. However, a detailed analysis of deaths, intensive care unit transfers, and the role of telemetry on decision making and management may shed some light on this question.

First, half of the patients who died (10 of 20) requested not to be resuscitated, making any life-saving procedures inappropriate should a major arrhythmic event be discovered. Telemetry monitoring was discontinued for these patients after an initial period of observation. In addition, 1 patient died due to causes unrelated to the cardiovascular system, and another patient had end-stage heart failure. The remaining 8 patients died within 4 days of hospitalization. Telemetry monitoring had been discontinued before death in 5 of 8 of these patients. Although we cannot infer causality regarding the usefulness of telemetry in preventing death, it is clear that 15 of 20 of these patients did not benefit from telemetry.

Second, telemetry confirmed or identified a life-threatening arrhythmia that led to an acute intervention and transfer to an intensive care unit in 19 of 2,240 patients (0.8%). Telemetry may have helped avert an immediate death in some of these patients. The 4 patients with atrial fibrillation or flutter were transferred after attempts to control their rhythm had failed and not because they sustained an unexpected tachyarrhythmia with hemodynamic compromise. All other transfers were mandated by clinical assessment (88 patients) or routinely after an intervention (134 patients). Major arrhyth-

mic events were rare and telemetry monitoring did not play a major role in intensive care unit transfers.

Third, by understanding the role of telemetry on decision making, one could identify if any adverse outcome might have been prevented. This may be the case for any of the 127 patients (5.7%) in whom telemetry was perceived as useful but did not alter management, or for any of the 156 patients (7%) in whom telemetry findings led to modifications of management. However, we do not imply that changes in decision making will translate into better outcomes. It is unlikely that telemetry was anything more than reassuring for the remaining 1,760 patients (78.6%) in whom the physicians believed that telemetry did not guide management or decision making.

Can we identify patients in whom telemetry is useful or in whom telemetry could be safely withheld? Patients admitted with syncope and chest pain syndromes had lower rates of unexpected intensive care transfer, and most were unrelated to arrhythmic conditions. Telemetry was less valuable for decision making in patients with chest pain syndromes and coronary artery disease awaiting revascularization (<5%), and more valuable in patients with an initial diagnosis of arrhythmia (51%) (Table IV). Because of the low yield and the low impact of information obtained from telemetry, this study questions the routine use of telemetry in patients with low-risk chest pain syndromes, and stable patients with coronary artery disease awaiting revascularization. It may be reasonable to withhold routine telemetry monitoring in these patients in the absence of a randomized controlled trial to resolve the issue. In a recent consensus guideline, it has been suggested that low-risk patients with unstable angina can be safely evaluated further as outpatients.¹⁷

There were limitations to this study. First, selection bias may have occurred since mortality in 9% of patients not followed prospectively was significantly higher than in patients who were followed prospectively. This implies that these patients may have been sicker and the role of telemetry for decision making may have been greater. However, they were not transferred to an intensive care unit more frequently than patients followed prospectively. Therefore, we may be underestimating the usefulness of telemetry. Second, this is an observational study and no causal inferences can be made. A randomized controlled study is needed to further address the impact of non-intensive care telemetry units. Third, observation bias may have occurred by the cardiologists recording the telemetry findings. No information was obtained directly from the arrhythmia detection instruments to assess this. Our findings do not apply to patients recovering from cardiac revascularization, or patients transferred from an intensive care unit after a cardiac event.

In conclusion, cardiologists perceived telemetry as a useful adjunct to care in 12.6% of admissions to a non-intensive care telemetry unit. In most cases, the usefulness of telemetry was subtle, either by reassuring the physician or altering medications. Less than 1% of all patients admitted to the telemetry unit had arrhythmias detected by telemetry that led to an intensive care unit transfer.

Future studies should include cost-effectiveness analysis and multivariate modeling of prognostic factors to identify patients who may benefit from telemetry.

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IMPLICATIONS FOR COST-EFFECTIVENESS

Cardiac arrhythmia monitoring in non-intensive care units is common and adds to health care costs both directly (as a direct charge to the patient) and indirectly (time spent and expense generated by physicians, nurses, and technicians monitoring and responding to cardiac rhythms). In our study, arrhythmias detected by telemetry led to intensive care unit transfers in <1% of all admissions and 0.3% of patients with chest pain syndromes. Telemetry was perceived to be of marginal value to the clinicians in only 4% of the latter group.

It is reasonable to withhold routine telemetry monitoring in patients with low-risk chest pain syndromes until a randomized controlled trial resolves the issue.

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